

Kostadin Devedzhiev

☎ +44 7950 483811 | ✉ kostadin.g.devedzhiev@gmail.com | 🌐 kostadindev.github.io

SUMMARY

PhD student at the University of Cambridge (CHIA, Trustworthy AI Lab) researching **human–AI orchestration**. Previously three years at Stellar Cyber building AI security products shown at RSA and Black Hat.

EDUCATION

University of Cambridge

Oct 2026 – Present

PhD in Human-Inspired Artificial Intelligence · Queens' College

Cambridge, UK

- **Human–AI orchestration:** multi-agent systems in which AI agents and people work together under varying expertise, cost, and availability.
- Centre for Human-Inspired AI (CHIA), Trustworthy AI Lab; co-supervised by Prof. Umang Bhatt and Prof. Adrian Weller

University of Cambridge

Oct 2025 – Jun 2026

MPhil in Human-Inspired Artificial Intelligence · Homerton College

Cambridge, UK

- **Awarded with Distinction** · 82.6 / 100
- Centre for Human-Inspired AI (CHIA), Trustworthy AI Lab; supervised by Prof. Umang Bhatt
- Thesis: *Ad Hoc Human–AI Orchestration via Skill Inference*

Stony Brook University

Aug 2018 – May 2022

BS in Computer Science & Applied Mathematics and Statistics

Stony Brook, NY

- Specialisation: AI & Data Science · CS Honors Program · GPA 3.89 / 4.00 · Summa Cum Laude

University of Hawai'i at Hilo

Aug 2019 – May 2020

National Student Exchange Program · Computer Science

Hilo, HI

- GPA 3.97 / 4.00

RESEARCH EXPERIENCE

University of Cambridge

Jul 2026 – Sep 2026

Research Assistant, Centre for Human-Inspired AI (CHIA)

Cambridge, UK

- Research on **Hybrid Workforces**, a collaboration between CHIA and **Accenture Singapore**.
- Developing skill inference, the Human API, and orchestration controls.

Stony Brook University

Aug 2021 – May 2022

NLP Research Assistant

Stony Brook, NY

- Designed **Recursive QA**, a framework for generating formal-representation annotations of natural-language specifications via guided question-answering over constituency parse trees.
- Reduced redundant annotation options using affinity clustering on Levenshtein distance; achieved **>80% inter-annotator agreement** with experienced users completing annotations in ~30 s.
- Built a full-stack annotation tool with account management, work-history tracking, and graph visualisations.

University of Hawai'i at Hilo

Jun 2020 – Aug 2020

AI Research Assistant

Hilo, HI

- Developed an IoU / F1-based threshold estimation heuristic to optimise real-time CNN proposal acceptance for **invasive-species detection** in drone imagery.
- Enhanced a human-in-the-loop semi-automatic annotation tool that progressively increases AI assistance as model accuracy improves during training.

PROFESSIONAL EXPERIENCE

Stellar Cyber

Software Engineer

May 2022 – Sep 2025

San Jose, CA

- Architected a **multi-agent autonomous SOC system** that triages security cases by correlating alerts with LLMs and generating analyst-ready summaries; featured as Stellar's flagship demo at **RSA 2025** and **Black Hat 2025**.
- Built the chat interface, session management, and multi-chart visualisation suite for **Open XDR Investigator**, a GenAI copilot translating natural-language prompts into Elasticsearch queries; headlined **RSA 2024** and **Black Hat 2024**.
- Implemented bidirectional WebSocket communication with parallel data and LLM requests, **cutting average response times by 70%** and enabling real-time progress streaming.
- Developed knowledge-enriched GPT agents for log analysis, product metrics, connector normalisation, and data-source classification, each capable of visualisations, API calls, and script execution.
- Owned the product analytics framework (Mixpanel), producing reports that informed UX redesigns, storage optimisation, and behavioural segmentation.
- Maintained **>90% test coverage** on owned components, **reducing bug filings by 34%** year-over-year.

University of Hawai'i at Hilo

Creative Electronic Media Assistant

Mar 2020 – May 2020

Hilo, HI

- Built a hologram display API and developed the Data Visualisation Lab website.

Vivansa

Software Engineering Intern

Jun 2019 – Aug 2019

Sofia, Bulgaria

- Improved CRM front-end components and resolved erroneous database entries through root-cause analysis and automated data-cleaning procedures.

RESEARCH & PROJECTS

Ad Hoc Human–AI Orchestration via Skill Inference – MPhil Thesis

- Models each teammate's ability as a **Bayesian belief over a taxonomy of skills**. Because the skills are correlated, measuring one also updates the others, so profiles converge on fewer observations than scoring each skill separately.
- Released as **skillinfer**, an open-source Python package on PyPI, and previewed in Tailor.

Tailor – Human-AI Orchestration Platform

- Visual workflow builder that combines AI agents with human review steps and four levels of governance control, built for regulated industries.

PUBLICATION

P-M Binder, **K. G. Devedzhiev**, A. T. Runyan. *Motional emf generated by squeezing an elliptical conducting loop*. European Journal of Physics, 2020. [doi:10.1088/1361-6404/ABB066](https://doi.org/10.1088/1361-6404/ABB066)

TALKS & PRESENTATIONS

Oxford Human–Algorithm Interaction (HAI) Workshop 2026, Saïd Business School, Oxford (Jul 2026) – Talk: *Ad Hoc Human–AI Orchestration via Skill Inference*.

CHIA Annual Conference 2026: AI for Good, Cambridge Union (Jun 2026) – Poster: *Ad Hoc Human–AI Orchestration via Skill Inference*.

ICO Technical Workshop: Know Your Data, Information Commissioner's Office, London (Mar 2026) – Participant; AI transparency and personal data in foundation-model training.

TEACHING & MENTORSHIP

AI Leadership Mentor, UniHawk (Kuwait City, Dec 2025) – Mentored high-school students through a week-long AI app-building bootcamp.

Teaching Assistant, Stony Brook University (Aug 2020 – May 2021) – AMS 210: Applied Linear Algebra; held weekly office hours and graded coursework.

Linear Algebra Grader, University of Hawai'i at Hilo (Mar 2020 – May 2020) – MATH 311: Linear Algebra.

Computer Science Grader, University of Hawai'i at Hilo (Oct 2019 – Dec 2019) – CS 150: Introduction to Computer Science.

CERTIFICATIONS

IBM Generative AI Engineering – IBM

SKILLS

Domains	Machine Learning, NLP, Human-Computer Interaction, Data Science, Full-Stack Development
Languages	Python, JavaScript, TypeScript, C
ML & AI	PyTorch, Hugging Face, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, Plotly
Web	React, Angular, Tailwind, FastAPI, Node.js, Express, Flask
Data & Infra	PostgreSQL, MongoDB, Elasticsearch, Pinecone, Redis, Docker